Astronomy A2: Light, Telescopes, and Stars

MAIS Physics & Astronomy Club (PAC)

Unlike laboratory physicists, astronomers cannot touch a star, put it on a scale, or insert a thermometer into its photosphere. Virtually everything we know about stars—their temperatures, radii, compositions, and lifetimes—comes from decoding the starlight collected by telescopes.

1. Luminosity, Flux, Magnitudes, and Telescopes

A star of intrinsic power (luminosity) **L** radiates isotropically across a sphere of area $4\pi\mathbf{d}^2$ at distance **d**, delivering energy per unit area per unit time (flux) $\mathbf{F} = \mathbf{L}/(4\pi\mathbf{d}^2)$. Historically, Hipparchus ranked visible stars into six brightness classes; Pogson formalized this logarithmic scale so that a difference of 5 apparent magnitudes corresponds to a flux ratio of exactly 100:

$$\mathbf{m}_1 - \mathbf{m}_2 = -2.5 \log_{10} ((\mathbf{F}_1)/ (\mathbf{F}_2)). \tag{1}$$

A circular telescope objective of aperture diameter **D** has two primary jobs: collecting light over area $\pi(\mathbf{D}/2)^2 \propto \mathbf{D}^2$, and resolving fine angular detail down to the wave-optics Rayleigh diffraction limit $\theta_{min} = 1.22\ \lambda/\mathbf{D}$ (in radians).

[1] Problem 1 (Why aperture rules). Compare a 200 mm (8-inch) club reflector telescope with a dark-adapted human eye of pupil diameter $\mathbf{d_e} = 7.0$ mm at visible wavelength $\lambda = 550$ nm.

(a) How many times more light does the telescope collect than a single eye? If the unaided eye can see stars down to apparent magnitude $\mathbf{m} = 6.0$ under a dark sky, how faint a star (in magnitudes) can the telescope reach?

(b) Compute the theoretical diffraction limit θ_{min} of the 200 mm telescope in arcseconds. Why do ground-based telescopes rarely achieve much sharper raw images than ~1″ without adaptive optics?

2. Thermal Radiation and Stellar Radii

To a good approximation, a star's photosphere radiates like an ideal thermal blackbody at effective surface temperature **T**. Two fundamental laws tie color and power directly to temperature:

$$\lambda_{max}\,\mathbf{T} = \mathbf{b} \approx 2.90 \times 10^{-3}\ \text{m·K},\ \mathbf{L} = 4\pi\mathbf{R}^2\,\sigma\mathbf{T}^4, \tag{2}$$

where Wien's displacement law gives the peak wavelength λ_{max} and the Stefan–Boltzmann law ($\sigma = 5.67 \times 10^{-8}\ \text{W·m}^{-2}\text{·K}^{-4}$) gives total radiated luminosity from surface area $4\pi\mathbf{R}^2$.

Remark. Avoid plugging in σ whenever you can! Dividing $\mathbf{L} = 4\pi\mathbf{R}^2\sigma\mathbf{T}^4$ by the solar values gives $(\mathbf{L})/(\mathbf{L}_\odot) = ((\mathbf{R})/(\mathbf{R}_\odot))^2\,((\mathbf{T})/(\mathbf{T}_\odot))^4$, which eliminates 4π and σ in a single line.

[2] Problem 2 (Measuring a red supergiant without resolving it). Betelgeuse in Orion has luminosity $\mathbf{L} \approx 1.0 \times 10^5\ \mathbf{L}_\odot$, and its continuous spectrum peaks in the near-infrared at $\lambda_{max} \approx 830$ nm (take $\mathbf{T}_\odot = 5{,}780$ K and 1 AU $\approx 215\ \mathbf{R}_\odot$).

(a) Find the effective surface temperature **T** of Betelgeuse.

(b) Estimate Betelgeuse's radius in units of $\mathbf{R}_\odot$ and in AU. Which planets' orbits would lie inside Betelgeuse if it replaced the Sun?

[2] Problem 3 (Planetary equilibrium temperatures). A rapidly rotating spherical planet of radius $\mathbf{R_p}$ orbits a star of radius $\mathbf{R}_\star$ and temperature $\mathbf{T}_\star$ at distance **a**. A fraction **A** (the Bond albedo) of incident starlight is reflected back to space, while the remaining fraction $(1 - \mathbf{A})$ is absorbed over cross-sectional disk area $\pi\mathbf{R_p}^2$ and re-radiated thermally over full spherical area $4\pi\mathbf{R_p}^2$.

(a) Show that the planet's radiative equilibrium temperature is $\mathbf{T_{eq}} = \mathbf{T}_\star\,(1 - \mathbf{A})^{1/4}\,\sqrt{((\mathbf{R}_\star)/(2\mathbf{a}))}$, independent of $\mathbf{R_p}$.

(b) Evaluate $\mathbf{T_{eq}}$ (in K and °C) for Earth ($\mathbf{A} = 0.30$, $\mathbf{T}_\odot = 5{,}780$ K, $\mathbf{R}_\odot = 6.96 \times 10^8$ m, $\mathbf{a} = 1.50 \times 10^{11}$ m), and explain why Earth's actual global mean surface temperature is +15 °C (288 K).

3. Nuclear Fuel and Stellar Lifetimes

During its main-sequence life, a star fuses hydrogen into helium in its core, releasing roughly $\varepsilon \approx 0.007$ (0.7%) of the fused rest mass as energy: $\mathbf{E} \approx \varepsilon\,\mathbf{m}_{fused}\mathbf{c}^2$. Equivalently, if $\Delta\mathbf{m}$ is the actual mass defect, $\mathbf{E} = \Delta\mathbf{m}\mathbf{c}^2$.

[3] Problem 4 (Live fast, die young). Assume a star can fuse the innermost 10% of its total mass **M** before exhausting core hydrogen.

(a) Using $\mathbf{M}_\odot = 1.99 \times 10^{30}$ kg and $\mathbf{L}_\odot = 3.83 \times 10^{26}$ W, estimate the Sun's main-sequence lifetime $\tau_\odot$ in billions of years (Gyr).

(b) Observations of main-sequence stars show the empirical mass–luminosity scaling $\mathbf{L} \propto \mathbf{M}^{3.5}$. Derive how main-sequence lifetime τ_{MS} scales with mass **M**, and estimate τ_{MS} for a bright blue 10 $\mathbf{M}_\odot$ star and for a dim 0.20 $\mathbf{M}_\odot$ red dwarf.